AIM60003 · AI-BASED MATHEMATICAL DISCOVERY · FALL 2026

Doing Mathematics with AI

Syllabus · Discussion · Visualization Project · Math Project

Sang-hyun Kim · KIAS School of Mathematics · KAIST AI4Math

SYLLABUS · ALL LECTURES 10:30-12:30 · IN ENGLISH

Four Lectures

1	**Oct 13 (Tue)**	Instructor's lecture	Thinking about Thinking · two visualizations · Thompson's group F
—	**Mid term**	No class	
2	**Nov 3 (Tue)**	Class discussion	Five topics on AI and the future of mathematics · five groups of five
3	**Dec 1 (Tue)**	Visual / Math projects I	Visualization (5 min / each) and math (5 min / each) projects
4	**Dec 3 (Thu)**	Visual / Math projects II	Visualization (5 min / each) and math (5 min / each) projects

Assessment and Deadlines

COMPONENTS

Discussion presentation	Group of 5 · Lecture 2	25%
Visualization project	Pair · Lecture 3 or 4	30%
Math project	Pair · Lecture 3 or 4	30%
Participation	Individual · Lectures 2–4	15%

25 students: one three-member team for each project type · same or different partners for the two projects · submissions via KLMS

Agenda

1	**Class dates**	Lecture 2: Nov 3 (Tue) · Lecture 3: Dec 1 (Tue) Lecture 4: Dec 3 (Thu) · all lectures 10:30–12:30
2	**Five groups of five: discussion topics**	Email me three choices by Oct 20
3	**Oct 20 · Proposal**	Discussion 1 page · visualization 1 page · math research 1 page
4	**Nov 1 · Demo**	Discussion presentation · visualization demo · math research demo You may upgrade the demo till the date of the presentation
5	**Dec 10 · Report**	Summary of the prompts · structure of AI interactions · lessons that you learned during the process

LECTURE 2 · NOV 3 · 20 MIN PER TOPIC: 10 PRESENTATION + 10 DISCUSSION

Five Discussion Topics

10:35	1	**AI and Mathematics as a Scientific Revolution**	A revolution? Which paradigms — and what is lost?
10:55	2	**The Qualities of Future Mathematicians**	Which qualities gain value? What survives of what we love?
11:15	3	**The Future of the Mathematical Community**	Who leads, who gets credit, who controls compute?
11:35	4	**The Art of New Mathematics**	Which skills matter when AI searches, computes, proves?
11:55	5	**The Ethics of AI-Assisted Research**	Who is an author? Who answers for errors?

LECTURE 2 · GROUPS OF FIVE · 5 MIN · UNDER 10 SLIDES

Discussion: Five Short Talks

What are facts and terms?

Why does it matter now?

Traditional position?

Oppositional position?

Seed questions, open questions, decision-now questions?

Prediction and value?

A fact, a prediction, or a value?

Reading List

YEAR	READING	WHY READ IT NOW	TOPICS
1936	**Turing**, "On Computable Numbers"	Computability and undecidability	1, 4
c. 1951	**Turing**, "Intelligent Machinery, A Heretical Theory"	Learning machines may outstrip us	1, 2
1951	**Gödel**, Gibbs Lecture: "Some Basic Theorems …"	Gödel's dilemma on mind and machine	2, 4
1956	**Wittgenstein**, *Remarks on the Foundations of Mathematics*	Surveyability; proof vs. calculation	1, 4
1962	**Kuhn**, *The Structure of Scientific Revolutions*	Paradigms and what revolutions lose	1
1992	**Smullyan**, *Gödel's Incompleteness Theorems*	Incompleteness, rigorous and compact	1
1993	**Jaffe and Quinn**, "Theoretical Mathematics"	Labeling speculation; rules for credit	3, 5
1994	**Thurston**, "On Proof and Progress in Mathematics"	Mathematics as human understanding	2, 3, 4
2017	**Stewart**, "Counting infinity", *Infinity: A Very Short Introduction*	Cantor: opposed at first, then loved	1, 2
2019	**Sutton**, "The Bitter Lesson"	Scaling beats built-in human knowledge	1, 3, 4

Topics: the five discussion topics of Lecture 2 · full references in the syllabus

Mathematical Aesthetics

YEAR	READING	WHY READ IT NOW
1908	**Poincaré**, “Mathematical Creation”, in *Science and Method*	The aesthetic sense as the sieve of discovery
1940	**Hardy**, *A Mathematician’s Apology*	Beauty, seriousness and permanence
1947	**von Neumann**, “The Mathematician”	A driving force, and the risk of drifting from empirical sources
1976	**Lakatos**, *Proofs and Refutations*	How definitions and rigor evolve through counterexamples
1997	**Rota**, “The Phenomenology of Mathematical Beauty”	Beauty as the experience of enlightenment

Visualization Project

Aim	A piece of mathematics made visible: a definition, a theorem, an example or a process
Audience	Chosen by the pair and stated in the proposal
Medium	A self-contained HTML page, no installation · a notebook or video when better suited
Examples	Fourier and Diffusion
Submissions	Proposal (1 page) Oct 20 · demo Nov 1, upgradable till the presentation · report Dec 10
Evaluation	Correctness · clarity for the audience · design and craft · honest account of AI's part

What can it show that a proof or a static figure cannot?

Visualization: Note and Demo

ONE-PAGE NOTE

1 Goal — what someone should see or understand

2 Mathematical content — what it carries and omits

3 Target audience — who, and what they already know

4 Prompts and steps — order, failures, fixes

5 Food for thought — AI's part, human judgment

FOUR-MINUTE DEMO

Q1 To whom is it addressed?

Q2 What should be delivered?

Q3 Why a visualization — and not a proof or a figure?

Q4 What are the aesthetic criteria?

Q5 What was the main technical difficulty, and the way past it?

Math Project Candidates

TYPE	SUCCESS LOOKS LIKE	MAIN RISK
Open problem from a list	Proof, counterexample, or sharper partial result	Already solved · false novelty
Question from a recent paper	An answer, a special case, or a reduction	Misreading the question
Own research problem	A lemma or step toward the thesis problem	Confidentiality
Known theorem, new route	A new route, compared with the literature	Known proof, unannounced
Computation or formalization	Data, conjecture, counterexample, Lean proof	Output without understanding

Type chosen in the proposal (Oct 20) · fallback within the same problem

MATH PROJECT NOTE: 1 PAGE + SLIDES

Design Questionnaire

Goal description	Solve this? Simplify this? Make a breakthrough toward this?
Initial prompt	Start research?
Continuation prompt	Please continue?
Staggering alert / reaction	If no advances for five times?
Context overflow / reaction	Write a hand-off for the next chat window?
Verification criteria	If the solution passes three independent agent reviewers' checks?
Check points	Report minor / major progresses?
Literature / human inputs	Thoroughly search the literature for similar results?

MATH PROJECT NOTE: 1 PAGE + SLIDES · EXAMPLE

Design Questionnaire

GOAL DESCRIPTION

Prove $PL_+(S^1)$ contains $\mathbb{Z}^2 * \mathbb{Z}$ by this construction

$\Rightarrow$ Prove $PL_+(S^1)$ does not contain $\mathbb{Z}^2 * \mathbb{Z}$

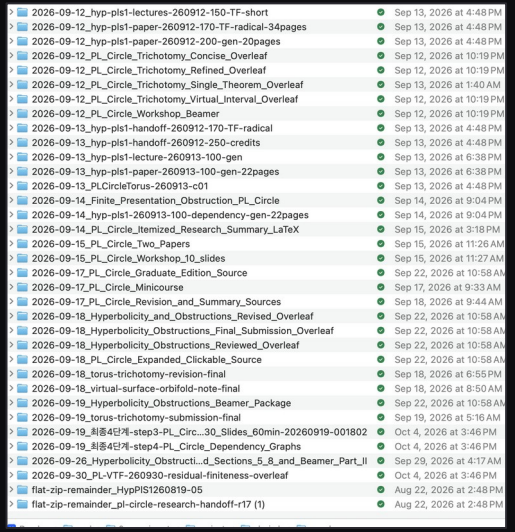
$\Rightarrow$ Classify finitely generated subgroups of $PL_+(S^1)$

INITIAL PROMPT

Give a rigorous solution to this mathematical research problem.

The solution should be at a publishable level.





SYLLABUS · COPE · AMS · LEIDEN DECLARATION (2026)

AI Use and Disclosure

Disclosure	Model, version, summary of prompts (not raw)
Novelty and credit	Literature search
Contribution	Who wrote which
Copyright	No unpublished work of others in commercial tools without permission

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